

Tau Core: An Atemporal Morphological Readout Framework for Emergent Physical Dynamics

Foundation Paper I

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Draft v0.1 — 2026

Abstract

Tau Core is proposed here as a cautious foundation framework rather than a completed physical theory. Its starting point is the possibility that ordinary dynamical physics is not fundamentally time evolution on a primitive spacetime, but an effective readout of a deeper atemporal response structure. The paper introduces the minimal formal spine

$$s \longmapsto R_\tau(s) \longmapsto U_i(R_\tau(s)),$$

where s is a parent-side seed, $R_\tau(s)$ is an endpoint-blind Tau response, and U_i are sector readouts such as spacetime, observer-time, quantum, gravitational, or mass/source readouts. The central claim is architectural: the formalism gives a language for separating definitions, postulates, conditional implications, toy mechanisms, and future empirical tests. The manuscript distinguishes the framework from block-universe ontology and from ordinary spacetime field theory, defines the Tau-side morphological body, formulates seven foundation postulates, introduces readout-relative hidden defects via relift failure, and gives finite toy models for hiddenness and effective time. It also records the main blockers: the paper does not derive general relativity, quantum mechanics, the Born rule, quantum field theory, the Standard Model, dark matter, dark energy, or a physical parent action. It is a position paper and formal conjecture scaffold, not empirical evidence for a new physical law.

Manuscript Roadmap

1	Purpose and claim boundary	3
2	Reader roadmap	3
3	Why a foundation paper is needed	4
4	The name Tau Core	5
5	Difference from a block universe	5
6	Difference from ordinary spacetime field theory	6
7	Core vocabulary	6
8	Minimal readout calculus	7
9	Nulls, quotients, and protected identity	8
10	Endpoint blindness as a scientific rule	9
11	How effective dynamics appears	10
12	Toy model with effective time	10
13	Seven foundation postulates	11
14	The claim ladder	12
15	A finite toy model	13
16	Morphological body versus projected morphology	14
17	Observer time	15
18	Quantum readout	15
19	Gravity and dark-sector readouts	16
20	Cosmological readout	16
21	Relationship to the existing paper sequence	17
22	Comparison with adjacent ideas	17
23	What would count as progress	18
24	What would count as failure	18
25	Likely objections	19
26	Minimal mathematical maturity checklist	19
27	The Roman foundation sequence	20
28	A compact formal summary	21
29	What is not proved	21
30	Conclusion	22

A. Minimal Tau Core spine

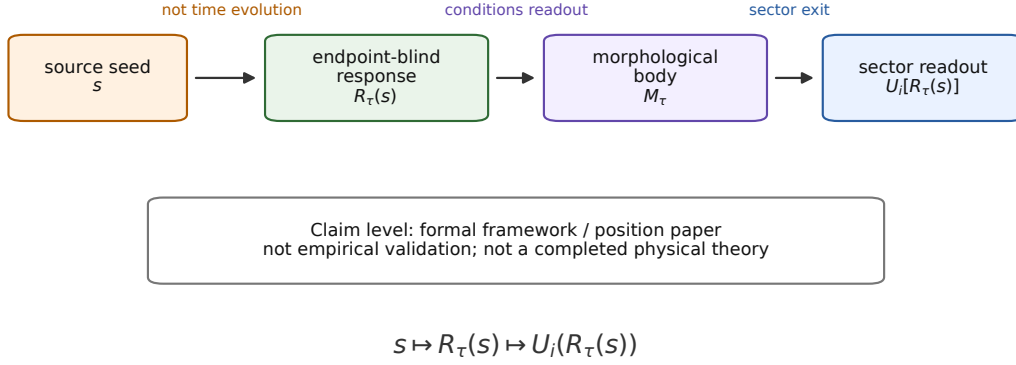


Figure 1: The minimal Tau Core spine used in Foundation Paper I. A parent-side seed s induces an endpoint-blind Tau response $R_\tau(s)$. A Tau-side morphological body M_τ conditions sector readouts U_i . This is a formal architecture, not empirical validation.

1 Purpose and claim boundary

This manuscript is the first Roman-numbered Tau Core foundation paper. It is separate from the Arabic-numbered Tau Core paper sequence, where the latter is used for bounded empirical or protocol packages. The purpose here is more basic: to state the current foundation language in a form that can be criticized, compared, and refined.

The claim boundary is deliberately narrow. The paper proposes a formal architecture. It does not claim empirical validation. It does not derive general relativity. It does not derive quantum mechanics. It does not replace Λ CDM. It does not solve the measurement problem, the dark sector, neutrino physics, or the Standard Model. It gives a vocabulary in which those later questions can be stated without assuming that spacetime time evolution is the primitive level.

The central idea can be written in one line:

$$s \mapsto R_\tau(s) \mapsto U_i(R_\tau(s)). \quad (1)$$

The first arrow is not time evolution. It is a parent-side response relation. The second arrow is a readout. A physical theory appears only after a readout has supplied a codomain, a null policy, a closure test, and a validation boundary.

This paper is therefore closer to a formal conjecture paper or a position paper than to a phenomenology paper. It is useful only if it makes the research program more falsifiable later. A broad ontology that cannot be turned into frozen source rules, null tests, or endpoint-blind predictions is not useful as physics. The framework below is written with that constraint in mind.

2 Reader roadmap

The paper has two jobs. First, it fixes a vocabulary for later Tau Core work. Second, it states what that vocabulary does not yet prove. The following roadmap is included to prevent the foundation language from being read as a completed physical model.

Part of the paper	Role	Claim level
Opening sections	Motivation, name, and relation to block-universe and spacetime-field views	Positioning
Core vocabulary	Parent response, morphological body, readout, relift, defect	Definitions
Formal spine	Minimal calculus, source discipline, effective time, and postulates	Framework postulates
Toy layer	Finite hidden-defect model and morphology/observer-time clarifications	Toy mechanisms
Boundary layer	Quantum, gravity, cosmology, relation to earlier papers, and objections	Boundary setting
Audit layer	Maturity checklist, foundation sequence, compact summary, and non-results	Audit discipline

Thus the paper should be read from weak to strong claim levels. Definitions do not become assumptions; assumptions do not become derived laws; toy models do not become empirical validation. Later papers may try to promote specific branches, but Foundation Paper I keeps those promotions outside its claim boundary.

3 Why a foundation paper is needed

Several themes in modern foundations already question whether time, spacetime, or local dynamics should be primitive. Canonical quantum gravity contains the well-known problem of time, where a timeless constraint equation can coexist with effective dynamics [5, 16]. Relational time constructions show how dynamics can be described through correlations rather than external evolution [12]. Timeless or configuration-space views have also been explored philosophically and mathematically [2]. Quantum reconstruction programs show that large parts of quantum formalism can be organized by operational principles rather than simply postulated [7, 4, 9].

Tau Core does not claim to solve those programs. It uses a different organizing question:

Can observed sector descriptions be treated as readouts of a deeper parent response, so that time evolution, quantum state, gravity, mass, and observer-time are effective sector descriptions rather than primitive categories?

This question is intentionally broader than any one branch. If it is too broad, it becomes metaphysics. The purpose of the present paper is to prevent that drift by specifying the minimal objects that must exist before a Tau Core statement becomes mathematically meaningful.

The immediate motivation is the following tension. Many physical descriptions use variables that are defined only after an observer, a section, a clock, a measurement basis, or a spacetime background is in place. But the same description often treats those

variables as if they were primitive. Tau Core instead separates:

parent response from sector readout.

That separation is the entire point of Eq. (1).

4 The name Tau Core

The name “Tau Core” is historical. The program began from a time-based intuition: perhaps the time seen by a four-dimensional observer is not a primitive background, but a projected response. This explains the use of τ , a symbol often associated with time or proper time in physics.

In the present architecture, however, τ does not mean ordinary time. It names the parent-state side of the theory. Time is one possible readout from that side. The distinction is important:

Symbol or phrase	Meaning here	Not claimed here
τ	parent-state side	ordinary time itself
X_τ	parent Tau state	detected physical field
$R_\tau(s)$	endpoint-blind response	time evolution
U_{time}	time readout	derived clock theory
M_τ	morphological body	observed 4D shape

Thus the theory is time-motivated but not time-reductionist. It does not say that everything is made of time. It says that time belongs to the readout side, while τ names the deeper side from which time and other physical sectors may be read.

5 Difference from a block universe

A block-universe reading treats a four-dimensional spacetime history as the final object. The comparison target is not a straw man. The relativistic background begins with Minkowski’s four-dimensional spacetime formulation [11]. The Rietdijk–Putnam–Maxwell line of argument then uses relativity of simultaneity to motivate four-dimensionalism or eternalism [15, 14, 10, 19]. Standard surveys of the philosophy of spacetime and time describe the resulting debates around being, becoming, eternalism, and the block universe [17, 8, 13, 18]. In that setting one may summarize the block-universe reading as

$$\text{Reality} = M_{4D}. \quad (2)$$

Tau Core instead treats a four-dimensional block, when such a block is a good description, as a readout:

$$M_{4D} = U_{4D}(R_\tau(s)). \quad (3)$$

The difference is not cosmetic. In a block-universe view the four-dimensional object is the final arena. In Tau Core it is an arena produced by a readout. The parent response can in principle have distinctions that one readout identifies, another ignores, and a third translates into effective dynamics.

This distinction also clarifies why Tau Core is not simply another form of eternalism. It can be compatible with an effective 4D block where that description works, but it does

B. Block-universe ontology versus Tau Core readout

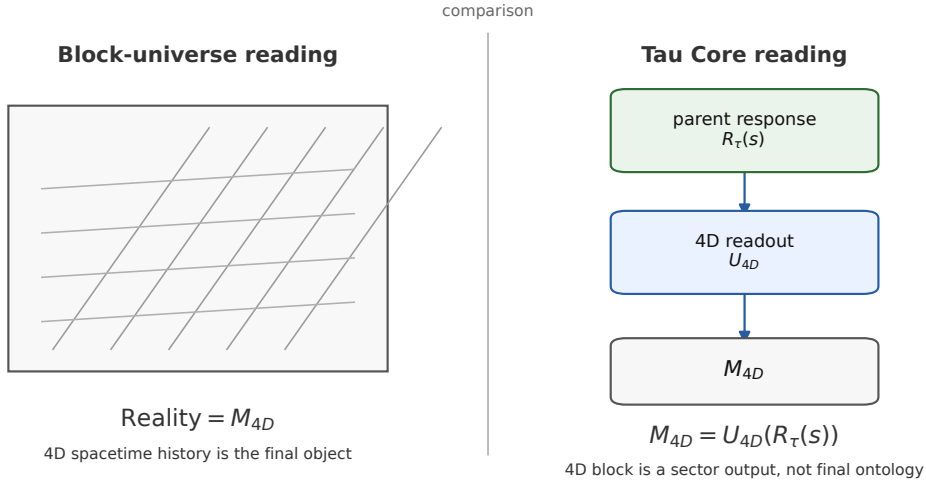


Figure 2: The block-universe comparison used in this paper. The left side represents the standard four-dimensional block reading, in which the spacetime history is the final object. The right side represents the Tau Core position: the same effective 4D block may be a stable readout $M_{4D} = U_{4D}(R_\tau(s))$ of a deeper parent response. The figure is a conceptual distinction, not a proof against eternalism.

not take the block as ontologically final. Nor does the paper refute the block-universe literature. It relocates the four-dimensional block from final ontology to sector output:

$$\text{block as final object} \longrightarrow \text{block as } U_{4D}\text{-readout.}$$

6 Difference from ordinary spacetime field theory

An ordinary field theory begins with a field on spacetime:

$$\phi : M_{4D} \rightarrow V, \quad (4)$$

and then studies the dynamics of $\phi(x, t)$. Tau Core rewrites the order:

$$\phi_{\text{obs}}(x, t) = U_\phi(R_\tau(s)). \quad (5)$$

The observed field is a readout. The spacetime coordinates are also part of the readout structure. This does not make ordinary field theory wrong. It makes it effective: a very successful description after a particular sector readout has become stable.

This is why Tau Core cannot claim physical success merely by writing a new readout map. To become physics, a readout must recover known limits and must survive empirical controls. If U_ϕ is arbitrary, the framework is empty. The research program therefore needs source-frozen readout rules, quotient-safe null policies, and explicit failure conditions.

7 Core vocabulary

Definition 1 (Parent seed). *A parent seed $s \in \mathcal{S}$ is a Tau-side source datum. It is not an observed endpoint and must not be chosen by looking at the endpoint residual that a later readout is supposed to explain.*

Definition 2 (Endpoint-blind Tau response). *An endpoint-blind Tau response is a map or relation*

$$R_\tau : \mathcal{S} \rightarrow \mathcal{R}_\tau$$

whose input is the parent seed and admissible parent-side constraints, not the observational endpoint. Endpoint blindness is a non-leakage rule, not a proof that the response is physically real.

Definition 3 (Morphological body). *The Tau-side morphological body M_τ is the intrinsic structured body that conditions readouts. It is not an observed 4D morphology label. It may carry sector, branch, wall, null, Hessian-support, admissibility, history, and observer/path-relevant structure. A projected morphology is only a readout handle on this deeper body.*

Definition 4 (Readout). *A readout is a sector map*

$$U_i : \mathcal{R}_\tau \rightarrow \mathcal{O}_i.$$

The codomain $\mathcal{O}_i$ may be a spacetime description, a time ordering, a quantum access state, a gravitational response, a mass/source record, or a more specialized observable structure. Each readout has its own null directions and validation boundary.

Definition 5 (Relift). *A relift for a readout U_i is a map or partial map*

$$J_i : \mathcal{O}_i \rightarrow \mathcal{R}_\tau$$

that attempts to reconstruct a parent-response representative from a readout record. It need not exist globally and need not be unique.

Definition 6 (Readout defect). *Given $r = R_\tau(s)$, a readout defect is*

$$h_i(r) = J_i(U_i(r)) - r,$$

whenever the subtraction is defined in a chosen linearized or quotient representation. More generally, the defect is the equivalence class of failure of $J_i U_i(r)$ to return the original parent-response representative.

Equation (6) is not a dark-matter formula or a quantum-collapse formula. It is an abstract incompleteness record. A physical branch must later specify the codomain, quotient, units, known limits, and empirical test.

8 Minimal readout calculus

The formalism above is intentionally weak. A readout is not yet a functor, a field equation, a probability rule, or a variational principle. Nevertheless, some minimal calculus is needed if the framework is to avoid being only language.

Let $\mathcal{R}_\tau$ be a parent-response space and let U_i be a readout. The first mathematical object induced by a readout is its indistinguishability relation:

$$r \sim_i r' \iff U_i(r) = U_i(r'). \quad (6)$$

This relation is readout-relative. A distinction can be invisible to one sector and visible to another. In this sense, a readout is not just a map to observables; it is also a rule for what counts as the same state for that sector.

Definition 7 (Readout kernel). *The readout kernel of U_i is the equivalence relation $\sim_i$ defined by Eq. (6). If $\mathcal{R}_\tau$ has a linear structure and U_i is linear, this reduces to the usual null space $\text{Ker } U_i$. In general it is an equivalence relation, not necessarily a linear subspace.*

Definition 8 (Readout refinement). *A readout U_j refines U_i if*

$$r \sim_j r' \implies r \sim_i r'.$$

Equivalently, U_j distinguishes at least as much parent-response information as U_i .

This gives a minimal partial order on readouts. A more informative readout is not automatically more physical. It may be harder to stabilize, harder to observe, or more vulnerable to noise. But the refinement relation prevents one from treating all readouts as interchangeable.

Definition 9 (Joint readout). *Given readouts U_i and U_j , the joint readout is*

$$U_{ij}(r) = (U_i(r), U_j(r)).$$

Its equivalence relation is the intersection:

$$\sim_{ij} = \sim_i \cap \sim_j.$$

Proposition 1 (Joint readouts refine their components). *The joint readout U_{ij} refines both U_i and U_j .*

Proof. If $r \sim_{ij} r'$, then $U_i(r) = U_i(r')$ and $U_j(r) = U_j(r')$. Hence $r \sim_i r'$ and $r \sim_j r'$. Therefore the joint equivalence relation is contained in each component equivalence relation. $\square$

This trivial observation is one reason multi-readout tests are valuable. If two independent readout families point to the same parent-side distinction, the candidate source class becomes more constrained. But this also introduces a warning: if the readouts are not independent, the joint readout may double count the same source coordinate. A mature Tau Core branch therefore needs a source-overlap ledger.

9 Nulls, quotients, and protected identity

Every physical readout has null directions: changes in the parent response that the readout treats as physically irrelevant. Some nulls are ordinary gauge redundancies. Some are observational limitations. Some are protected identity relations: distinctions that should not be counted as physical differences in a declared sector.

Let N_i denote the null relation for readout U_i . The quotient object is written

$$[r]_i \in \mathcal{R}_\tau / N_i.$$

The readout is quotient-safe only if

$$r N_i r' \implies U_i(r) = U_i(r').$$

If Eq. (9) fails, the readout assigns different observables to parent states that it also declares physically equivalent. Such a readout is internally inconsistent.

The converse is not required. A readout can identify more states than the declared null relation identifies. That is precisely readout loss. The important issue is whether the loss is allowed. In empirical branches this is handled by a source-freeze discipline: the readout-relevant source coordinates must be declared before endpoint scoring. In the foundation branch this becomes a source-priority problem: the selected readout should not erase protected source identity unless the erased distinction is explicitly routed to the null quotient.

Definition 10 (Protected source identity). *A protected source identity is an equivalence relation Q_{src} on parent-side seeds or source representatives that the theory treats as source-level identity. A readout rule is source-faithful if it does not collapse distinct Q_{src} -classes into one selected source class without declaring the collapse as protected null structure.*

This definition is not an extra empirical claim. It is a consistency demand. Without it, a model can hide arbitrary distinctions in the parent space and recover them only after seeing an endpoint. That would make the theory unfalsifiable.

10 Endpoint blindness as a scientific rule

Endpoint blindness is the operational face of the foundation idea. A readout model is endpoint-blind when the source, kernel family, amplitude rule, and null policy are fixed before the target endpoint residual is inspected.

This is not a philosophical preference. It is the only way a broad readout framework can avoid becoming curve fitting. If a galaxy rotation residual, a cosmological distance residual, or a quantum outcome is used to choose the source class that then explains it, the model has leaked the answer into its input.

The minimal endpoint-blind record contains:

- (E1) a source object or source proxy;
- (E2) a declared readout family U_i ;
- (E3) a null and quotient policy;
- (E4) a source-overlap decision for all channels used together;
- (E5) a freeze artifact or equivalent timestamped declaration;
- (E6) forbidden endpoint fields;
- (E7) controls and failure conditions.

Foundation Paper I does not run such an endpoint test. It states why later branches need it. The philosophical content of Tau Core becomes scientific only when it produces source-frozen rules that can fail.

11 How effective dynamics appears

Tau Core replaces primitive time evolution by readout-parametrized variation. Let $r = R_\tau(s)$ be atemporal at the parent-response level. Suppose a readout U_i produces a record that can be coordinatized by a parameter t_i :

$$U_i(r) = o_i(t_i).$$

Then the derivative

$$\frac{do_i}{dt_i}$$

is meaningful inside the readout. It need not correspond to a parent-time derivative. This is the minimal sense in which dynamics is emergent.

The distinction can be made sharper by considering two readouts of the same parent response. One readout may admit a temporal parameter t_A , while another admits a static relational record:

$$U_A(r) = o_A(t_A), \quad U_B(r) = C_B.$$

There is no contradiction. The first readout organizes the parent response as a time-indexed sequence. The second organizes it as a correlation or constraint. A conventional physical theory may then choose the first description because it is useful, local, and predictive. Tau Core's claim is weaker: usefulness of a temporal parameter does not make time primitive.

This is also why observer time must be sector-typed. A clock used in a galaxy-rotation analysis, a lensing time-delay analysis, a cosmological distance analysis, and a quantum measurement protocol are not automatically the same parent object. They may share a local proper-time limit, but that limit has to be earned.

12 Toy model with effective time

The finite model above can be extended to show effective time. Let

$$r = (a_0, a_1, \dots, a_n, c) \in \mathbb{R}^{n+2}.$$

Define a readout

$$U_{\text{seq}}(r) = (a_0, a_1, \dots, a_n),$$

and choose the index k as an effective time coordinate. The readout sees a sequence a_k . A finite-difference law can be defined:

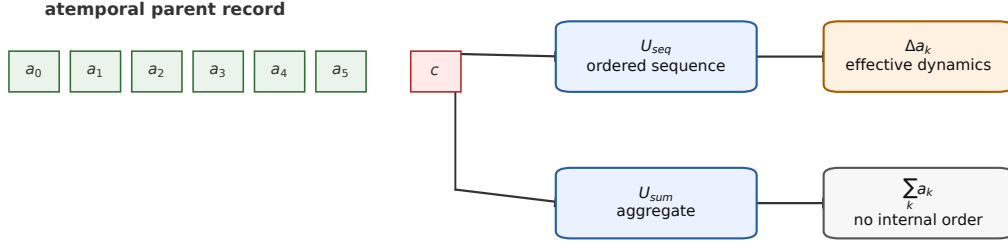
$$\Delta a_k = a_{k+1} - a_k.$$

Nothing in the parent object has evolved. The parent response already contains the indexed structure. The readout has supplied an ordering and therefore an effective dynamics.

Now define a second readout

$$U_{\text{sum}}(r) = \sum_{k=0}^n a_k.$$

E. Effective time as a readout ordering



The parent record need not evolve for a readout to display an ordered time parameter.

Figure 3: A finite toy model for effective time. The parent record is not itself evolving. The sequence readout U_{seq} exposes an ordered list and therefore supports finite differences Δa_k . The aggregate readout U_{sum} exposes a different observable and loses the internal ordering. Effective dynamics is therefore readout-relative in this model.

This readout sees no time sequence, only an aggregate. The hidden component relative to U_{sum} is the loss of ordered information. The hidden component relative to U_{seq} includes c . Again, hiddenness is readout-relative.

This toy model is deliberately too simple to be physics. It only shows that the sentence “time is a readout” can be made mathematically literal without introducing backward causation or a second physical clock.

13 Seven foundation postulates

The postulates below are not proved in this paper. They define the candidate framework. Later papers or theory notes may try to derive or weaken them.

Postulate 1 (Tau substrate). *There exists a parent-level structure $\mathcal{T}$ that is not, at the foundation level, ordinary spacetime evolution.*

Postulate 2 (Seed-response). *Physical configurations are represented at the parent level by endpoint-blind responses to seeds:*

$$s \in \mathcal{S}, \quad r = R_\tau(s) \in \mathcal{R}_\tau.$$

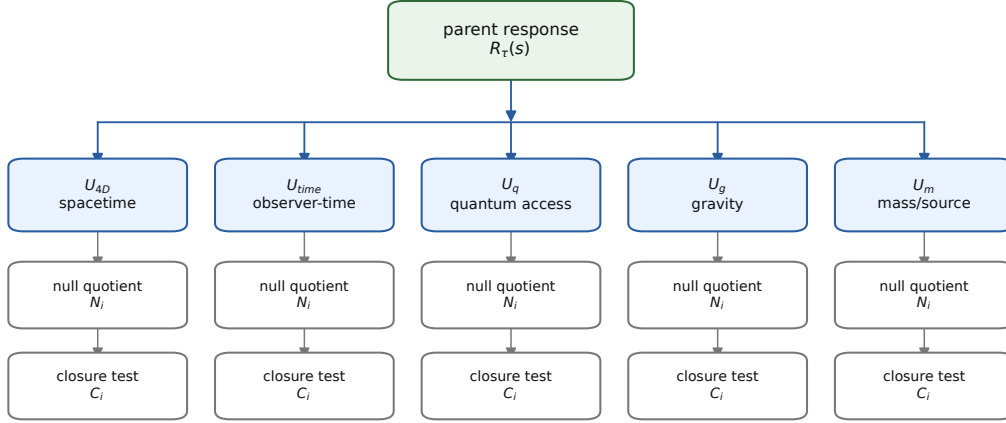
Postulate 3 (Endpoint blindness). *The response $R_\tau(s)$, and any source object used to select a readout kernel, must be specified without using the endpoint residual that the readout is later asked to explain.*

Postulate 4 (Readout emergence). *Observed physical descriptions are sector readouts:*

$$O_i = U_i(R_\tau(s)).$$

Time, spacetime, quantum access, gravity, and mass/source descriptions are readout candidates rather than primitive categories.

C. Readout atlas with null and closure boundaries



Each sector needs its own codomain, null quotient, closure test, and validation boundary.

A shared parent response does not imply automatic additivity of readout components.

Figure 4: A parent response can have several sector readouts. The same parent object may be read as spacetime, observer-time, quantum access, gravitational response, or mass/source structure. Each sector requires its own codomain, null policy, closure test, and validation boundary. The paper does not assume that these readouts are automatically additive or mutually complete.

Postulate 5 (Effective dynamics). *Time evolution appears only when a readout admits a stable ordering or clock coordinate. A derivative such as*

$$\frac{d}{dt}U_i(R_\tau(s))$$

is an effective readout derivative, not a primitive parent-time derivative.

Postulate 6 (Hidden defect). *If a readout and its allowed relift fail to recover the parent-response representative modulo the declared null relation, the non-reliftable component is recorded as a hidden defect of that readout.*

Postulate 7 (Tension as readout failure). *Some observational or theoretical tensions may be modeled as readout, closure, or relift defects. This is a research hypothesis. It becomes physical only after source-frozen rules, controls, known limits, and held-out tests are supplied.*

14 The claim ladder

A central discipline of Tau Core is to separate claim levels. The framework contains definitions, postulates, conjectures, conditional theorems, toy models, numerical evidence, and empirical validation. These are not interchangeable.

The most common failure mode for broad foundation programs is promotion by language: a metaphor becomes a mechanism, a mechanism becomes evidence, and evidence becomes validation. This paper avoids that promotion. For example, saying that observer-time is a readout does not derive a physical clock. Saying that dark-sector

F. Claim-boundary ladder

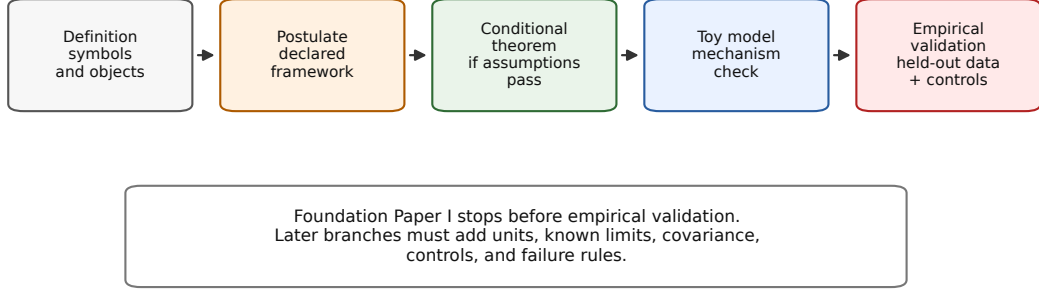


Figure 5: The claim ladder for Foundation Paper I. The present manuscript stops at definitions, postulates, conditional implications, and toy mechanisms. It does not claim empirical validation.

language may be a hidden-defect language does not remove dark matter. Saying that quantum state may be a readout does not derive Hilbert space.

Figure 6 gives the same discipline in a more compact form. Green entries indicate what this paper supplies as framework content. Orange and red entries indicate explicit non-claims and blockers. The point is not to make the paper look complete; it is to make incompleteness visible.

15 A finite toy model

The following model is intentionally small. Its role is not to reproduce physics. Its role is to show the readout logic without infinite formal machinery.

Let the parent response space be

$$\mathcal{R}_\tau = \mathbb{R}^3, \quad r = (a, b, c).$$

Let two readouts be

$$U_A(a, b, c) = (a, b), \quad U_B(a, b, c) = (a, c).$$

These readouts forget different coordinates. Choose canonical relifts

$$J_A(x, y) = (x, y, 0), \quad J_B(x, z) = (x, 0, z).$$

Then the readout defects are

$$h_A(r) = J_A U_A(r) - r = (0, 0, -c),$$

and

$$h_B(r) = J_B U_B(r) - r = (0, -b, 0).$$

The same parent response therefore has different hidden components under different readouts. There is no contradiction: each readout has a different null policy.

G. Foundation Paper I claim-status matrix

	Defined	Postulated	Toy mechanism	Validated	Open blocker
Tau substrate	yes	yes		not claimed	open
Seed-response map	yes	yes		not claimed	open
Morphological body	yes	yes		not claimed	open
Sector readouts	yes	yes	yes	not claimed	open
Effective time	yes	yes	yes	not claimed	open
Hidden defect	yes	yes	yes	not claimed	open
GR / quantum / QFT				not claimed	open
Empirical validation				not claimed	open

Green cells mark framework content supplied here; orange/red cells mark explicit non-claims or blockers.

Figure 6: Claim-status matrix for Foundation Paper I. The paper defines and postulates a parent-response/readout architecture and supplies toy mechanisms for effective time and hidden defects. It does not validate the framework empirically and does not derive GR, quantum mechanics, QFT, or a physical parent action.

Proposition 2 (Readout-relative hidden component). *In the toy model above, a component hidden under readout U_A can be visible under readout U_B , and conversely.*

Proof. The coordinate c is absent from $U_A(a, b, c)$ and appears as the defect $(0, 0, -c)$ under the canonical relift J_A . The same coordinate is present in $U_B(a, b, c)$. Similarly, b is visible under U_A and hidden under U_B . Thus hiddenness is readout-relative. $\square$

The lesson is small but important. A hidden component is not necessarily a new substance. It may be a component that a chosen readout cannot relift. Whether this idea has anything to do with dark sectors, quantum measurement, or cosmology is a later physical question. The toy model only licenses the logical grammar.

16 Morphological body versus projected morphology

The phrase “morphology” can be misleading if left unqualified. In ordinary astronomical use it often means visible shape or catalogue type. In Tau Core, the primary object is not the observed shape. It is the Tau-side morphological body M_τ .

The morphological body is the structured source object that conditions readouts. A galaxy image, a rotation curve, a cosmic-web tensor, a lensing map, or a quantum measurement context can be a projected handle on the body, but not the whole body. This distinction became necessary because a single 4D appearance can underdetermine the readout-relevant source. For example, a present projected shape may not contain enough information about history, observer/path geometry, closure class, or null structure.

For Foundation Paper I, the minimal role of M_τ is:

$$O_i = U_i[M_\tau](R_\tau(s)). \quad (7)$$

The bracket indicates that the readout map may depend on the morphological body. It does not say that M_τ has been physically derived. It says that readout selection is not source-free.

D. Relift failure as a readout-relative hidden defect

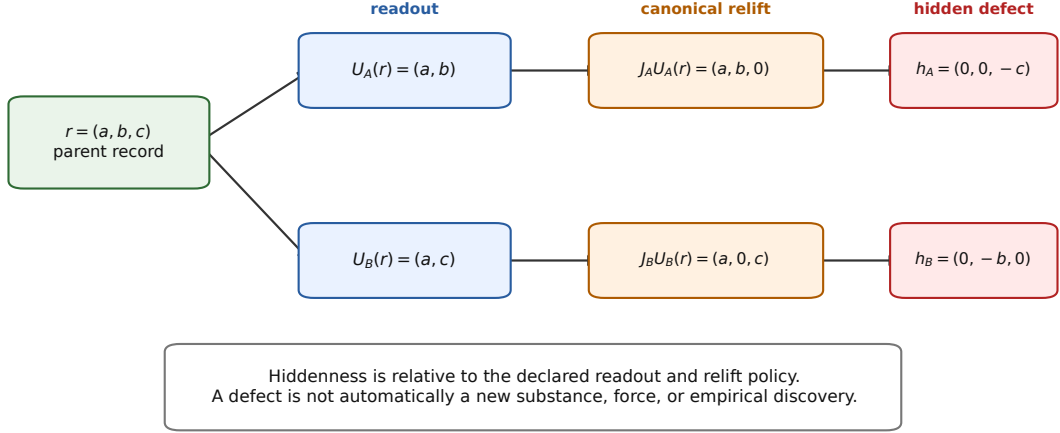


Figure 7: Readout-relative hidden defects. Two readouts of the same parent response can have different relift failures. A hidden component is not automatically a new force or substance; it is first an incompleteness record relative to a declared readout.

This point also prevents double counting. If two proposed correction terms use the same source coordinate, they should not be added independently unless a source-side split justifies the separation. Otherwise the readout model is only relabeling the same source information twice.

17 Observer time

Tau Core treats observer time as a readout, not as a primitive parameter on the parent side. The schematic form is

$$t_{\text{obs}}^S = U_{\text{time}}^S(R_\tau(s), M_\tau, \mathcal{O}_S, \gamma_S), \quad (8)$$

where S is a sector, $\mathcal{O}_S$ is observer-interface data, and γ_S denotes path or section data relevant to that sector.

Equation (8) is not a derived clock theory. It is a discipline rule. It says that an observer's time belongs to the observer's readout bundle. A later physical theory must explain when t_{obs}^S reduces to proper time, when it admits corrections, and how those corrections can be measured without circularity.

This also clarifies why future-directed or history-like terms in Tau Core should not be read as backward causation. They are components of a deeper parent response whose different 4D readout slices may appear as past, present, or future. The parent object is not evolving in an external time. The observed ordering is created inside the readout.

18 Quantum readout

The quantum branch is one of the natural targets for the readout language, but it is also one of the most dangerous places to overclaim. A minimal quantum readout might have

the form

$$q = U_q(R_\tau(s)), \quad (9)$$

where q is an access-state or probability-assignment record. But this is far below deriving Hilbert space or the Born rule.

Existing reconstruction programs show that quantum theory can be approached from information-theoretic or operational axioms. Examples include Hardy-style, purification-style, and local/reversible reconstruction routes [7, 4, 9]. Generalized probabilistic and Jordan-algebraic routes clarify how much structure is needed before Hilbert-space quantum theory is forced [3]. Tau Core can use these results as boundary discipline. It must not claim that a finite contextual readout grammar already implies complex Hilbert space.

The current safe statement is:

If Tau Core supplies an ordered convex access-state space, separating effects, an appropriate self-dualizing pairing, continuous homogeneous reversible symmetry, composition rigidity, and trace uniqueness, then a Born/Hilbert representation becomes a conditional target.

This is a conditional reconstruction route, not a derivation. The present paper includes quantum readout only to show where it sits in the common architecture.

19 Gravity and dark-sector readouts

The gravity branch motivates the hidden-defect vocabulary. A gravitational readout may fail to relift all source-relevant structure into a Newtonian, relativistic, or closure-compatible description. In abstract notation:

$$h_g = J_g U_g(R_\tau(s)) - R_\tau(s). \quad (10)$$

If a component of h_g survives the declared null quotient and has a measurable 4D readout, it becomes a candidate residual channel. This is still not a dark-matter replacement. It is a source of possible residual structure.

The same discipline applies to dark-sector language. Tau Core may eventually ask whether some dark-sector phenomena are better understood as readout-relative hidden defects. Foundation Paper I does not answer that question. It only defines the level at which the question becomes meaningful. To make a dark-sector claim, one would need source-frozen kernels, conventional baseline comparisons, covariance/systematics treatment, and failure criteria.

20 Cosmological readout

Cosmology is another natural readout target because the observed expansion history, distance ladder, growth data, lensing data, and CMB inference are all compressed through model-dependent readout assumptions. A Tau Core cosmological readout might be written

$$O_{\text{cosmo}} = U_{\text{cosmo}}(R_\tau(s), M_\tau^U, \gamma_{\text{path}}, \mathcal{O}_{\text{obs}}), \quad (11)$$

where M_τ^U is a universe-scale morphological body, γ_{path} is path/environment data, and $\mathcal{O}_{\text{obs}}$ is observer-interface data.

Again, the notation is not a result. It is a warning against compressing all cosmological tensions into a single function of the scale factor. A real cosmological Tau Core branch would need multi-probe covariance, source-native void/web/path/time channels, and a clear relation to Λ CDM limits.

21 Relationship to the existing paper sequence

The Arabic-numbered Tau Core papers are not required for the logic of this foundation paper. They are narrower, operational packages. Some examine galaxy residuals, some define source-frozen morphology or projection kernels, some formulate dark-sector decision protocols, and some define cosmological or web-witness audit ladders. They are evidence-seeking packages, not proofs of the parent ontology.

Foundation Paper I has the opposite role. It defines the vocabulary that those operational papers can later use. It should not borrow their numerical outcomes as proof. A positive empirical audit in a later branch can motivate the framework, but it does not validate the parent theory unless the path from parent source to readout rule is itself frozen, non-circular, and tested.

This separation is important for scientific criticism. A critic should be able to reject a galaxy-kernel result without thereby rejecting the formal readout architecture. Conversely, a critic can reject the foundation architecture while still accepting that a particular empirical paper performed a reproducible audit.

22 Comparison with adjacent ideas

Tau Core overlaps in spirit with several families of ideas, but it should not be confused with them.

Relational time. Relational-time approaches show how dynamics can be recovered through correlations. Tau Core agrees that external time need not be primitive, but it places this inside a broader readout atlas. Time is one readout among several, not the only reconstruction target.

Shape dynamics and relational configuration. Shape-dynamics-inspired work emphasizes that spacetime structure may be reformulated through different symmetry or relational choices [6]. Tau Core uses this as a cautionary analogy: one must state which structure is primitive and which is recovered. It does not import shape dynamics as a proof.

Quantum reconstruction. Quantum reconstruction programs provide a model for turning operational principles into formal structure. Tau Core’s quantum branch must meet a similar standard. It cannot derive Hilbert space merely by naming a quantum readout.

Holography and error correction. Holographic quantum error correction shows that bulk-like structure can be encoded and recovered nontrivially from boundary data [1]. Tau Core treats this as an analogy for readout/reconstruction discipline, not as evidence for the Tau ontology.

23 What would count as progress

The framework becomes stronger only if abstract readout language is converted into bounded tasks. Examples include:

- (1) deriving a non-circular parent response law R_τ ;
- (2) defining a source-realizable morphological body M_τ ;
- (3) proving that a readout map has a quotient-safe null policy;
- (4) deriving a known physical limit, such as proper time or Newtonian gravity, as a controlled readout limit;
- (5) producing source-frozen empirical rules that survive held-out tests;
- (6) preserving negative results by demoting failed kernels instead of retuning them after endpoint inspection.

The weakest useful form of progress is a toy model that exposes a blocker. A failed theorem can be valuable if it shows which extra principle is needed. For example, local convexity or a positive Hessian may not force global exposed-face duality; finite contextual grammars may not force Hilbert space; and source stability may not force a unique source basis. Such negative results help keep the framework honest.

24 What would count as failure

A reviewer should not have to infer the failure conditions. At the foundation level, Tau Core would be weakened or demoted if the following failure modes persist after reasonable attempts to repair definitions:

- (F1) **No source-realizable parent object.** If no non-circular parent-side source class can be specified for a branch, the branch remains vocabulary rather than physics.
- (F2) **No quotient-safe readout.** If declared null-equivalent parent representatives produce different observables, the readout is internally inconsistent.
- (F3) **No known-limit recovery.** If a branch cannot recover the relevant standard limit, such as proper time, Newtonian gravity, Born probabilities, or Λ CDM behavior where appropriate, it should not be promoted.
- (F4) **Endpoint leakage.** If the source class, kernel family, or amplitude rule is selected using the residual it later explains, the result is a fit, not a Tau Core derivation.
- (F5) **Uncontrolled rescue.** If every failure is repaired by adding an unconstrained projection channel after endpoint inspection, the framework loses scientific content.

These are not merely methodological preferences. They are the conditions under which the present foundation language can fail to become a physical research program. A negative result satisfying these conditions should be preserved as a real demotion, not hidden by changing terminology.

25 Likely objections

A foundation framework should be written so that criticism has handles. We therefore list several likely objections and the intended answer.

Objection 1: this is only a change of language. This objection is serious. The framework is only useful if readout language forces new constraints. The minimum non-linguistic constraints are endpoint blindness, null/quotient safety, source-overlap discipline, known-limit recovery, and explicit failure conditions. Without these, Tau Core would be a vocabulary rather than a research program.

Objection 2: the parent response is unobservable. The parent response is not directly observed by assumption. But this alone is not fatal; many theoretical objects are inferred through constrained effects. The relevant question is whether a declared readout rule produces predictions or exclusions that could have failed. If no such rule is produced, the parent response remains a metaphysical placeholder.

Objection 3: the framework can explain anything. This is the most important danger. Tau Core must therefore preserve negative results. A failed frozen kernel should be demoted, not retuned after looking at the residual. A wrong readout family should remain wrong. A source class that cannot be independently justified should not be promoted. If the framework always rescues itself by adding another projection, it fails as science.

Objection 4: atemporal response sounds like a block universe. The difference is Eq. (3). A block universe treats a 4D history as the final object. Tau Core treats a 4D history as one possible readout. The distinction matters only if other readouts impose independent constraints. If they do not, Tau Core collapses back into ordinary effective description.

Objection 5: no physical law is derived. Correct. Foundation Paper I does not derive a physical law. It defines the objects a later derivation must use. A future physical law would require a parent action or closure principle, a domain, a null policy, variation support, units, calibration, and known limits.

26 Minimal mathematical maturity checklist

Before a branch of Tau Core should be treated as more than a toy model, it should satisfy at least the following checklist.

Requirement	Meaning
Source declaration	The parent-side source or proxy is declared before endpoint scoring.
Readout codomain	The observable space $\mathcal{O}_i$ is specified.
Null policy	Gauge, redundancy, and protected-null directions are declared.
Quotient safety	Equivalent parent representatives give the same readout.
Relift policy	The branch states whether and how J_i is defined.
Known limit	The readout recovers a standard limit where required.
Failure rule	Failed controls cause demotion, not post-hoc retuning.

This checklist is intentionally weaker than a full physical theory. It is also stronger than a metaphor. A branch that cannot fill these rows should not be advertised as a physical result.

27 The Roman foundation sequence

This paper is the first member of a separate Roman-numbered foundation series. It should not be confused with the Arabic-numbered empirical/protocol sequence. The intended division is:

Paper 1–14 bounded empirical or protocol packages,
Foundation Paper I, II, ... conceptual and mathematical foundation papers.

The proposed next papers are:

- **Foundation Paper I:** atemporal morphological readout; framework only, with no physical validation claim.
- **Foundation Paper II:** quantum readout; no unconditional Hilbert, Born-rule, or QFT derivation claim.
- **Foundation Paper III:** hidden defects and dark sectors; no dark-matter or dark-energy replacement claim.
- **Foundation Paper IV:** observer-time and cosmology; no Hubble-tension or Λ CDM replacement claim.
- **Foundation Paper V:** flavor and neutrino readout; no neutrino-anomaly or Standard Model claim.

The purpose of this split is defensive in the good sense. A foundation paper can state a formal conjecture without pretending to be a data paper. A data paper can test a narrow rule without requiring acceptance of the full ontology. This separation makes both kinds of work easier to criticize.

28 A compact formal summary

The framework can be summarized as the following object package:

$$\mathfrak{T} = (\mathcal{S}, \mathcal{R}_\tau, R_\tau, M_\tau, \{U_i\}, \{N_i\}, \{J_i\}).$$

Here $\mathcal{S}$ is a seed/source domain, $\mathcal{R}_\tau$ is a response domain, R_τ is the endpoint-blind response map or relation, M_τ is the morphological body, U_i are readouts, N_i are null policies, and J_i are optional relifts.

The minimal consistency conditions are:

$$s \text{ is fixed without endpoint leakage,} \tag{12}$$

$$r = R_\tau(s) \text{ is defined on the declared source domain,} \tag{13}$$

$$rN_i r' \Rightarrow U_i(r) = U_i(r'), \tag{14}$$

$$h_i(r) = J_i U_i(r) - r \text{ is interpreted only modulo } N_i, \tag{15}$$

$$\text{known physical limits are recovered before physical promotion.} \tag{16}$$

The strongest claim of this paper is that $\mathfrak{T}$ is an internally organized starting package for a readout-based research program. This is not a formal consistency proof of all possible Tau Core branches, and it is not a proof that Nature uses such a package.

29 What is not proved

For clarity, we list the main non-results.

- The paper does not prove that the Tau substrate exists physically.
- The paper does not provide a parent action.
- The paper does not derive physical time.
- The paper does not derive general relativity.
- The paper does not derive quantum mechanics, the Born rule, collapse, or QFT.
- The paper does not derive the Standard Model.
- The paper does not explain neutrino physics.
- The paper does not eliminate dark matter or dark energy.
- The paper does not validate any empirical Tau Core claim.

These limitations are not footnotes. They are part of the claim. A foundation paper that hides its blockers is not useful. The goal is to make the blockers visible enough that later work can attack them one at a time.

30 Conclusion

Tau Core is presented here as an atemporal morphological readout framework. The basic proposal is that observed physical dynamics may be sector readouts of an endpoint-blind parent response:

$$s \mapsto R_\tau(s) \mapsto U_i(R_\tau(s)).$$

In this reading, a 4D spacetime block is not the final ontology; it is one readout:

$$M_{4D} = U_{4D}(R_\tau(s)).$$

Physical time is likewise a readout channel, not the meaning of τ itself.

The paper’s contribution is therefore not a new empirical result. It is a bounded vocabulary: parent seed, endpoint-blind response, morphological body, sector readout, relift, hidden defect, observer-time readout, and claim ladder. This vocabulary lets later work ask sharper questions. Which readouts are source-realizable? Which defects survive the quotient? Which known limits are recovered? Which source-frozen kernels survive held-out data? Which candidate branches fail?

The strongest honest summary is:

Tau Core is not offered here as a completed theory. It is offered as a formal research program in which time evolution, spacetime fields, quantum state, gravity, and dark-sector language can be re-expressed as readout problems over a deeper atemporal morphological response.

If that re-expression leads only to flexible terminology, it should be discarded. If it leads to frozen source rules, recovered limits, and failed or successful held-out tests, then the framework becomes a scientific program. This paper is the first step: it states the program in a form that can be criticized.

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